A120592 concise calculus certificate

VERIFIED

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Definition and first terms

$5\,A(x) = 4 + 4\,x + A(x)^3$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(2)*T(x); T=x+3*T^2+2*T^3.`

$a(0), \ldots, a(7) = 1, 2, 6, 40, 330, 3048, 30156, 312528.$

Kernel used throughout: $\rho(u) = u\,(1 - 3\,u^1 - 2\,u^2)$, so $x = \rho(T(x))$.

Typogeometry

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2 : 3,\ \Delta_3 : 2.$

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_2) \in \mathbb{Z}^2_{\geq 0} : \sum_{j=1}^{2} j k_j = n - 1\}, \quad K = \sum_{j=1}^{2} k_j,$$

$$a(n) = \frac{2}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \ldots, k_2} 3^{k_1} 2^{k_2}.$$

$$a(n) = \frac{2}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{2} P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$P_0(n) = -108\,n^2 + 12$$
$$P_1(n) = -216\,n^2 - 324\,n - 108$$
$$P_2(n) = 17\,n^2 + 51\,n + 34$$

Differential equation

$$\left(12\,x^2\right) A(x) + \left(-108\,x^3 - 108\,x^2\right) A'(x) + \left(-108\,x^4 - 216\,x^3 + 17\,x^2\right) A^{(2)}(x) = 0.$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{198}{17} & \frac{30}{17} & \frac{3}{17} \\ \frac{180}{17} & \frac{18}{17} & \frac{12}{17} \\ 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 & 0 \\ \frac{96}{17} & \frac{13}{17} & \frac{3}{17} \\ \frac{60}{17} & \frac{6}{17} & \frac{4}{17} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Certificate.

$$R(n, u) = \frac{-72\,n\,u^4 - 144\,n\,u^3 + 42\,n\,u^2 + 114\,n\,u - 17\,n - 24\,u^4 - 48\,u^3 - 6\,u^2 + 6\,u}{\rho(u)^1}.$$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))$.

Matrix dimensions: 6×6; remainder 2×3, rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.